

Key Factors Influencing Housing Affordability in New Zealand

Arjun Bhatt, Liam McKay, Phoebe Zhang

Data309: Data Science Capstone Project

October 19th, 2025

1. Introduction

1.1. Organization: Describe the organization and its domain

The topic area of the project was housing affordability in New Zealand, focusing on the Canterbury, Auckland and Wellington regions. The aim of the project was to determine the major socioeconomic, demographic, political and market drivers of housing affordability in the selected regions.

1.2. Goals: Document the goals/activities/research questions

The purpose of our project was to identify the main driving factors of housing affordability in the three largest regions of New Zealand (Canterbury, Auckland, Wellington). We also aimed to compare the factors between the regions (Canterbury, Auckland, Wellington) to see if they differed from region to region (urban areas vs. rural areas in each region). Finally, we also wanted to investigate how COVID-19 had changed the observed trends.

In a data science context, we wanted to conduct trend analysis spanning the pre-COVID, during-COVID, and post-COVID periods, model the interaction between key variables such as interest rates, income, building consents, and house prices, and discover the effects policy changes had on housing affordability. If we faced time constraints, our most important goals were to model housing affordability based on the variables we collected, investigate the evolution of income to housing cost ratios, and analyse the impact of government policies.

1.3. Constraints: Describe any constraints imposed by the organization

One obstacle we expected to encounter was data integration, as data may be in different formats from different sources. The reformatting and cleaning of the data were the first steps of the project, so addressing it early allowed us to avoid having errors later that could have been avoided by unifying formats.

All data used were either public or anonymised to ensure privacy. We had to extract data from PDFs and HTMLs manually to ensure that the data used were accurate, and integrity was maintained. GitHub was used to handle version control, track changes and collaborate. External internet APIs for data processing could not be used due to privacy concerns.

2. Data

The data were sourced from reputable New Zealand institutes, including Stats NZ, Reserve Bank of New Zealand (RBNZ), the Ministry of Housing and Urban Development (HUD), and Infometrics. These datasets were produced using official surveys, aggregated market data, and administrative data.

From Stats NZ, we collected information on household incomes by region, building consents, and dwelling and household estimates. This source covered a lot of regional information that was key for modelling. Our most important source was Data.govt, which covered affordability indices (mortgage, deposit, and rent), fixed and floating mortgage rates, and total housing stock. The affordability indices ended up playing a crucial role in uncovering what drove affordability issues. Figure.nz provided unemployment rates, infometrics gave us house prices by region, and finally RBNZ provided debt serving data and CPI. While these sources did not provide large volumes of data, they still contributed to painting a clearer picture of housing affordability trends.

We expected integrating multiple sources into a single dataset would be a significant challenge, as the same variables could contain inconsistent naming conventions. Missing data for some time periods or incomplete coverage for certain regions was another issue we needed to address. Some other problems we thought would occur were incompatible temporal granularity (e.g. certain datasets reported monthly, while others are available on a quarterly basis), CSV parsing errors due to encoding or column naming issues, and region name matching across datasets.

2.1. Descriptive statistics: Describe the quantity, nature and quality of each source of data

The dataset contained 901 observations of 24 variables. Each observation captured data for a New Zealand city on a specific year from 2015 to 2025. The other variables contained economic data averaged over the year, and additional geographic information such as region.

The dataset included both quantitative and qualitative variables. The quantitative variables were Log_price, representing the logarithm of the average house price; Income_level, which indicates the median household income; Unemployment_Rate, referring to the yearly unemployment rate. It also contained Mortgage_Index, measuring mortgage affordability; Deposit_Index, which measured deposit affordability; and Rent_Index, representing rent affordability; Consents, indicating regional building consents per 1000 people; and Year, which specifies the calendar year. The qualitative variables were Region, which categorized each observation by region (Auckland,

Canterbury, Wellington, or Other), and Urban_Rural, which separates observations into either urban or rural classification.

The data used for this project were sourced from New Zealand institutes such as Stats NZ, RBNZ (Reserve Bank of New Zealand), Data Govt and Infometrics. The datasets were gathered from official surveys, aggregate market data, and administrative data. This meant that all the data were of good quality with well-defined variables. The dataset was also pre-processed to remove missing and invalid values.

2.2. Missing: Describe data missingness (especially missing-not-at-Random)

The dataset was complete for almost all variables after the cleaning process. Building consents were a notable exception, as there was significant missingness. All regions were marked as NA for building consents in years before 2019. This meant about one third of the dataset was missing values for this variable. This meant the building consents variable was missing-not-at-random (MNAR), and that the missingness represents the lack of availability of the data before 2019, rather than an issue with the data quality itself.

The inclusion of this variable would have shrunk the sample size from 901 to 570 observations, meaning roughly 30% of our observations would have been removed. The missingness in this variable would impact the confidence in the findings, as a lower number of observations results in more variance in regression output. Because the missingness was not at random, imputation of the missing values would be inappropriate, as it would be generating values with no historical basis. For these reasons Consents_Per_1000 was excluded from the final regression models. No significant missingness was observed in the other variables after the cleaning process.

2.3. Outliers: Identify any significant univariate outliers and comment on data quality based on this

Outlier detection occurred using a value greater than or less than 1.5 times the interquartile range (IQR) value (see Table A1). Most of the variables had little to no outliers, which indicated a good quality of data. The exception was Consents_Per_1000, which was an expected high number of outliers due to the lack of data before 2019. The log transformation was suitable on the house price data, as no outliers were seen.

2.4. Imbalance: Acknowledge any classification class imbalance

As this project is less focused on classification, but rather on continuous variables and regression, class imbalance is not a large issue. However, there were more observations within the Auckland and Canterbury regions than Wellington and Other regions. This is unsurprising, as it is the result of the different population sizes of these regions. This imbalance was not expected to invalidate the results but was noted, as regions with fewer observations were likely to have less accurate estimates and larger confidence intervals.

2.5. Near-Zero-Variance: Acknowledge variables that are mostly constant

Most variables showed significant variation across observations, so near-zero variance was also not a major concern. All observations used in the final models (e.g., Income, Unemployment rate, and Affordability Indices) varied enough to provide meaningful contributions to the models.

A few variables such as CPI showed smaller variance over shorter time periods, which was expected as they were aggregated at the national level. However, these indicators still varied enough to capture long term trends related to housing affordability when looking across the full 2015-to-2025-time span.

3. Methodology

3.1. Data retrieval and cleaning: Document (in words rather than code) how the data was retrieved, wrangled and cleaned.

The final dataset used for this assignment was created by merging data from several official sources. Each source was downloaded in an uncleaned format and imported into the project GitHub repo, where the data was cleaned and merged in RStudio (see Table A2).

The first step of the cleaning process was to merge all the datasets within each source. This process was different for each source but generally consisted of the same steps. First, only the year of each data variable was extracted. Observations were then aggregated by year, using the average. Unnecessary variables were dropped, and any observations with missing or invalid values were removed. This created one dataset per source, with one observation per year – and where relevant – separate observations for each city.

The next step was to merge these five datasets into one final master dataset which would be used for modelling. This was performed by completing a full join on all datasets, joining by the year variable, resulting in a table with 1909 observations of 84 variables. The dataset had a lot of missingness, as some variables had data going back to as far as 1979, so more cleaning had to be performed.

The first cleaning step was to derive each region (Auckland, Canterbury, Wellington, Other), and its type (Urban or Rural). Region was derived by mutating the dataset to compare the observed area to a manually created area to region map. Type was derived by comparing the area to see if it ended with the string 'City' or 'District', with City considered to be Urban and District considered to be Rural.

The next step in the cleaning process was to address missing values. To do this, a plot was created to show the proportion of missing values for variable for each year (see figure A1). 2015 was chosen to be the cutoff year, as it provided the largest number of variables, whilst keeping the missingness low, as well as providing enough observations to compare pre-COVID, during- COVID, and post- COVID. This reduced the master dataset from 84 to 44 variables.

The next step in the cleaning process was to unpivot variables where possible. This was performed by combining regional variables into a single variable. For example, the uncleaned dataset had three variables Unemployment_Auckland, Unemployment_Christchurch, and Unemployment_Wellington. These were combined into a single variable 'Unemployment', which contained the unemployment rate for the region the observation was describing rather than recording all regions separately. This was repeated six times, resulting in the master dataset going from 44 to 24 variables. This significantly improved the readability of the dataset and allowed for these variables to be used during modelling.

The final step in the cleaning process was to rename each variable to further improve readability. This process was completed, and the complete cleaning process resulted in a master dataset comprised of data taken from five reputable New Zealand sources. These sources included official statistics on housing prices, mortgage and deposit rates, economic indices, and building consents.

3.2. Data Strategies: Document (in words rather than code) what special steps needed to be taken for outliers, missing values, imbalanced outcomes, high-cardinality nominal variables, near-zero-variance variables.

As the dataset was built using well-defined variables sourced from official sources, no complex data sampling or feature selections strategies were required. We instead focused on a careful data cleaning and aggregation process to ensure consistency across regions and years.

3.3. Data Ethics: Discuss what data ethics issues exist and any strategies to mitigate these.

The data collected for this study were taken from open, trustworthy sources; however, it was still necessary to address several ethical issues. The primary concern of the research involved the utilization of data which could be used to infer population level measures, such as income, unemployment rate, and housing access/affordability. Even though the dataset did not include any personally identifiable information (PII), efforts were still taken to ensure that findings would not unfairly represent any regions.

To minimize these risks, only summaries of information were taken at region level, rather than including individual or household data which could infringe upon privacy but would also still allow for an analysis of housing trends for each region. In addition, results were interpreted conservatively to prevent false claims being made about regions or populations.

Another ethical consideration of the study was data representation. The dataset consisted of variables from several different sources, each of which recorded the data at a different time point with slightly different variable definitions, and these inconsistencies could potentially skew the models. This was mitigated by sourcing all the data from trusted official sources, standardizing all variables, and handling missing values in a way that was well documented.

Finally, all data cleaning transformations and modelling were fully documented in a repository to ensure that the findings were reproducible and that all steps were completely transparent. The data was also not manipulated to achieve a certain outcome, and all findings were reported as objectively as possible.

3.4. Methods: Document the ML algorithms attempted. Identify the “best” method(s) and which performance metric(s) were used to assess this. Discuss any particular characteristics (e.g. tolerant of missing values, transparent etc.). Record the preprocessing steps optimised for the best method(s). Discuss feature extraction strategies.

Multiple linear regression was used to model the relationship between housing affordability and key predictors such as income, unemployment rate, and affordability indices. The focus of our project was on understanding how individual variables impacted affordability trends, rather than predictive performance. For this reason, advanced machine learning methods were not required.

4. Results

4.1. Explain: Show charts and discuss each one. Give each chart a number and title. Make sure axes are properly labelled.

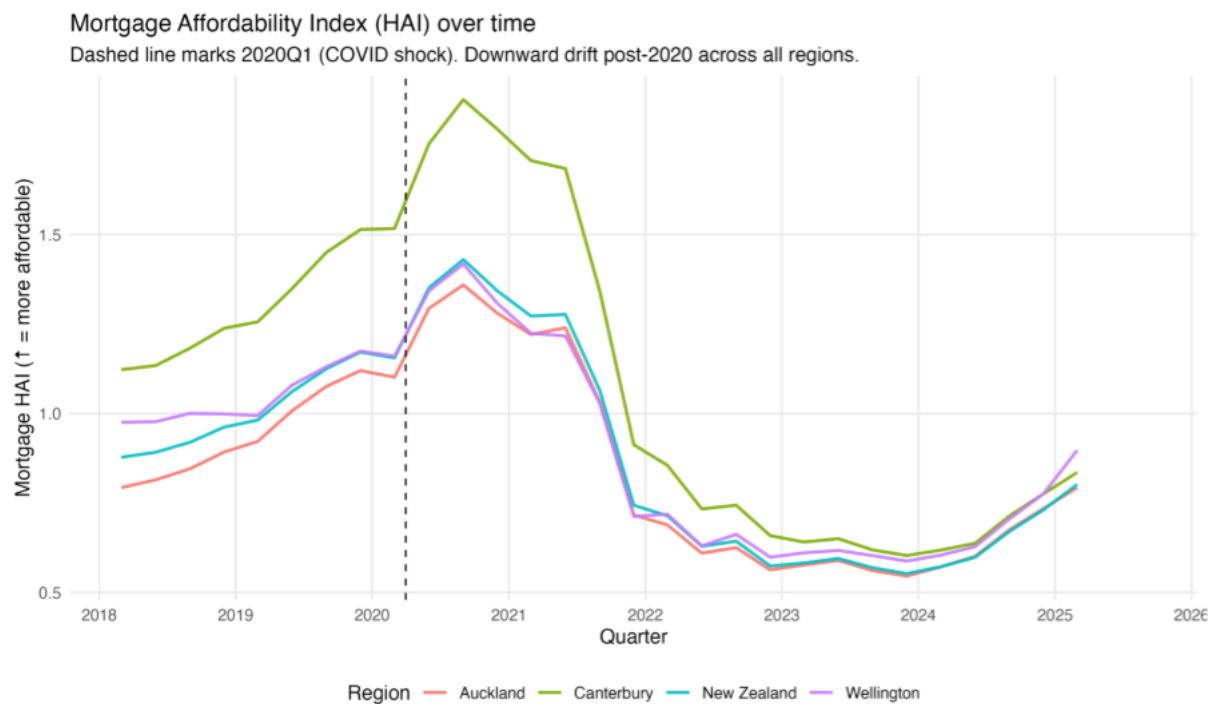


Figure 1. HAI Trend by Region (2018–2025)

File: 01_hai_trend.png Axes:

This figure showed the House Affordability Index (HAI) over time for Auckland, Canterbury, and Wellington. Affordability improved during the COVID-era lows in interest rates (2020–2021), then deteriorated sharply through 2022–2023 as rates rose, before stabilising modestly into 2025. Canterbury moved more sharply than the other regions, while Auckland and Wellington tended to track each other.

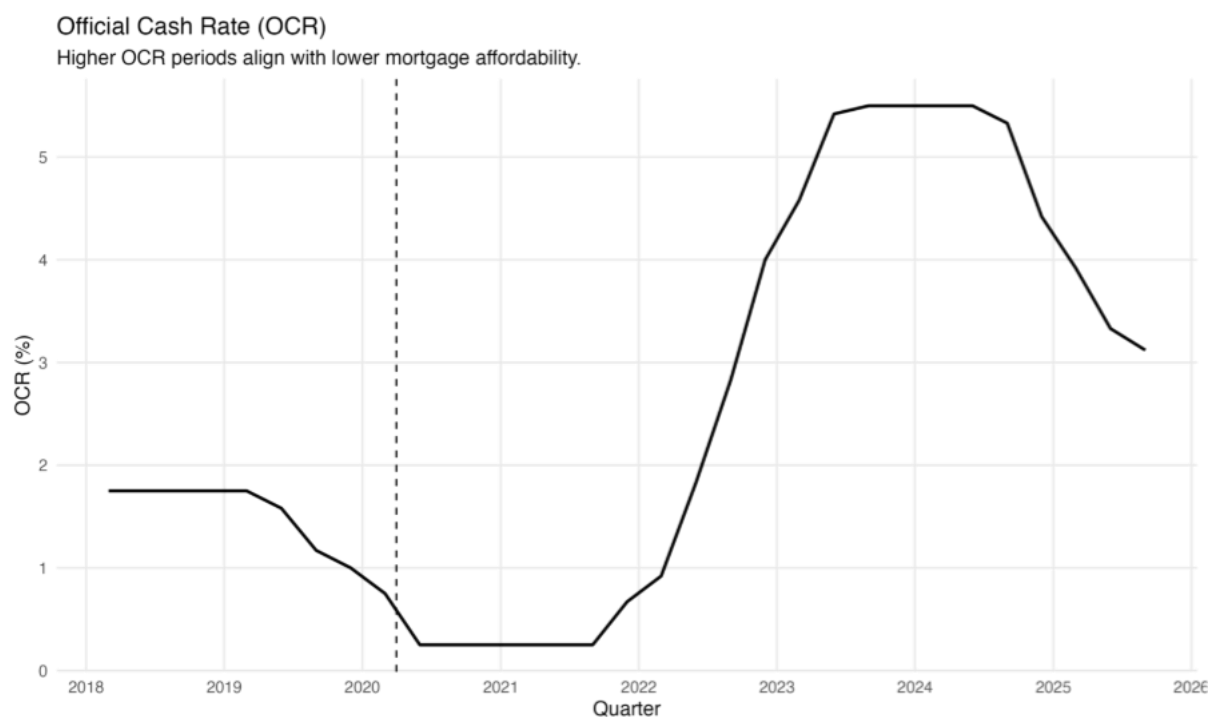


Figure 2. Official Cash Rate (OCR) over Time

File: 02_ocr_trend.png

This time-series displayed the Reserve Bank's policy rate falling to historic lows in 2020 and then rising quickly through 2022–2023. That tightening coincided with the period where our affordability measures worsened. The timing aligned closely with the HAI movements in Figure 1.

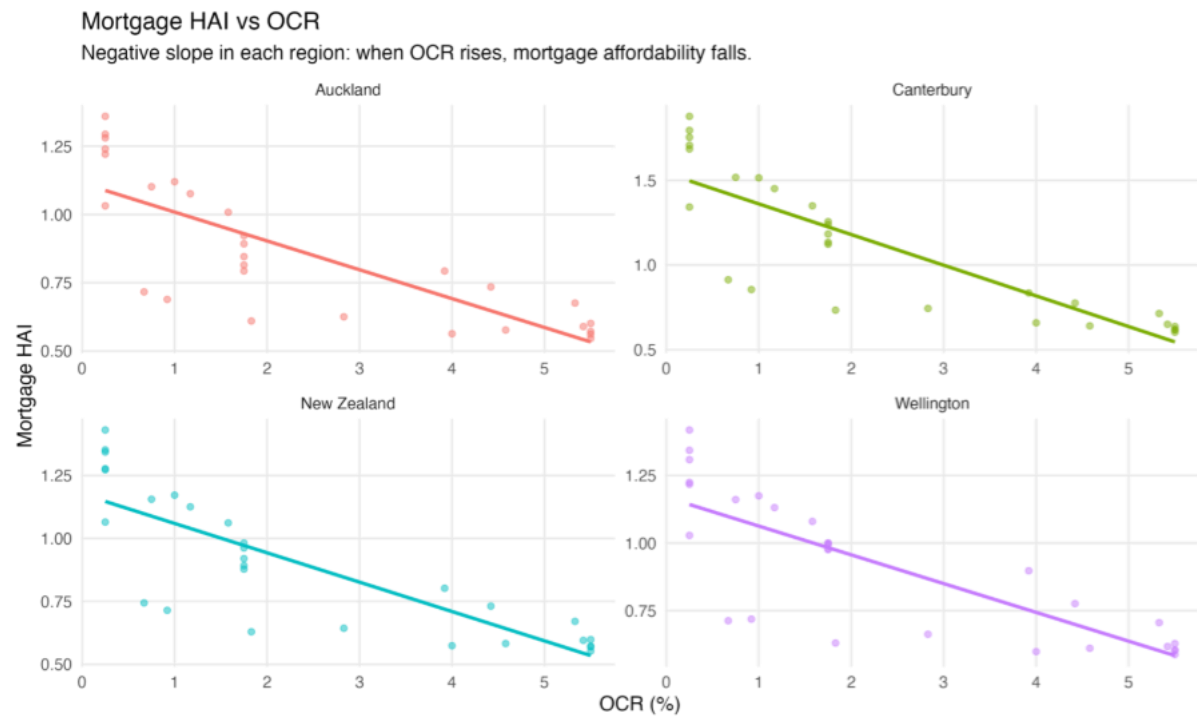


Figure 3. HAI vs OCR, by Region

File: 03_hai_vs_ocr_by_region.png

This scatter with fitted lines related affordability to the OCR for each region. The fitted lines sloped downward, indicating that higher policy rates were associated with lower affordability. The pattern appeared strongest in Canterbury and weaker, though still negative, in Wellington.

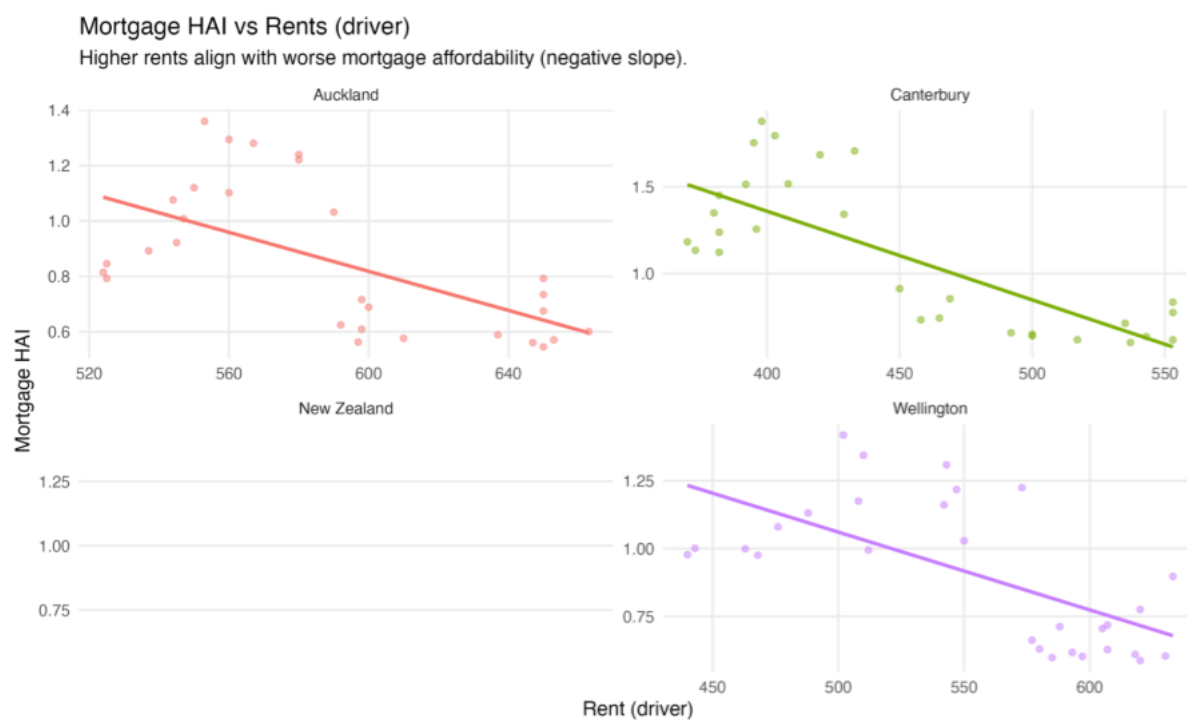


Figure 4. HAI vs Rent Index, by Region

File: 04_hai_vs_rent_by_region.png

This chart compared affordability with a rent burden index. As rent burden increased, the HAI tended to be lower, indicating worse overall affordability. The negative association was most visible in Auckland and Canterbury, with a weaker pattern in Wellington.

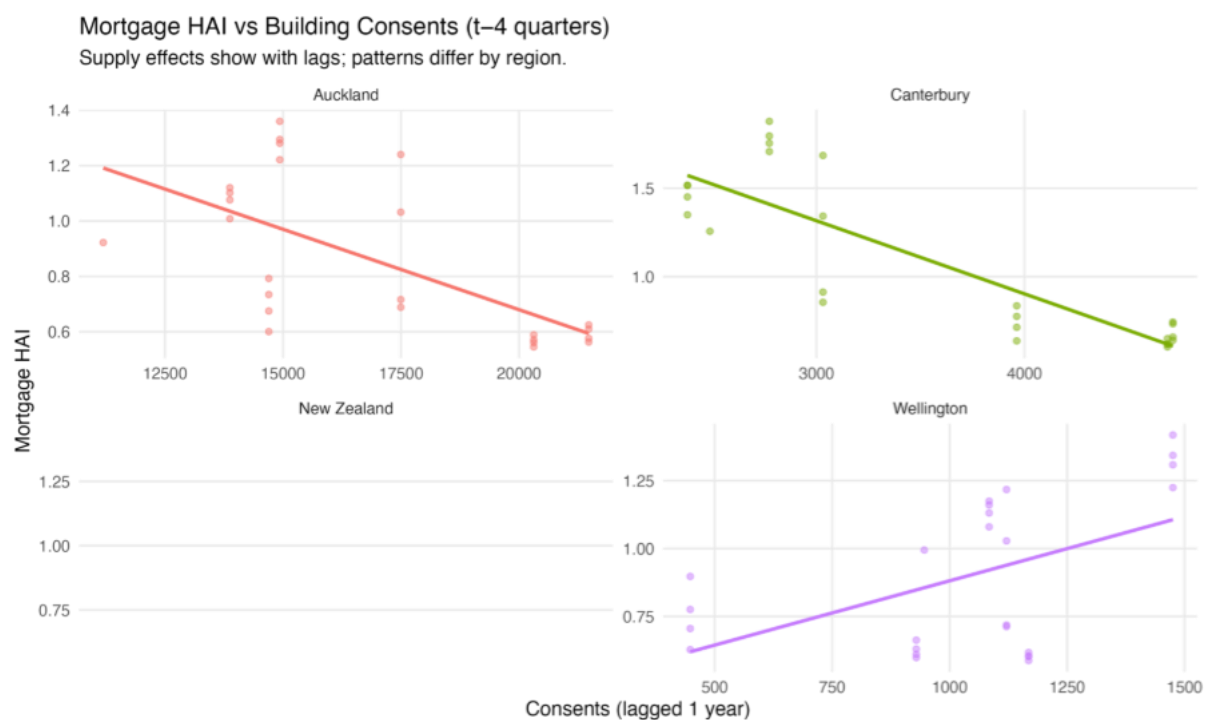


Figure 5. HAI vs Building Consents per 1,000 (lagged)

File: 05_hai_vs_consents_lag4_by_region.png

This plot examined the relationship between affordability and lagged building consents (a proxy for supply). The association was noisy and varied by region, but the direction was broadly consistent with the idea that more approved dwellings eventually supported affordability. Our regressions later echoed this with sign-sensitive, mixed effects.

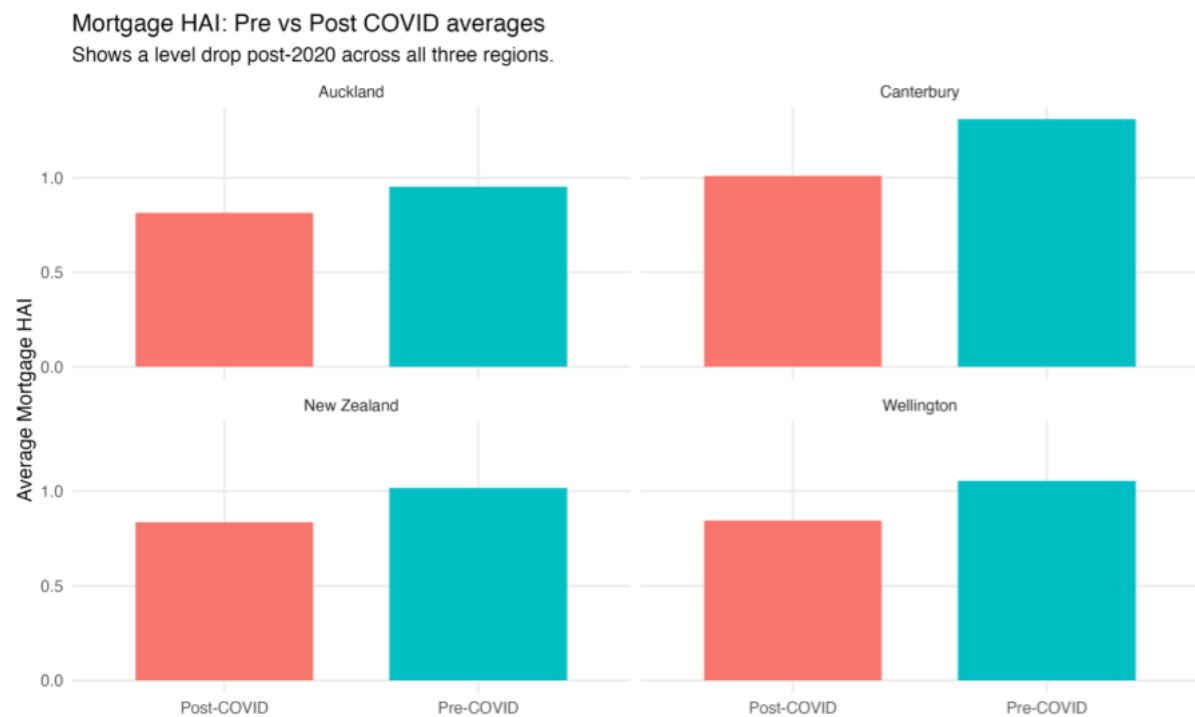


Figure 6. HAI: Pre- vs Post-COVID by Region

File: 06_prepost_covid_hai.png

This figure compared average affordability before and after 2020. Affordability was lower in the post-COVID period across all regions in our data. The drop aligned with the lift in borrowing costs and other post-pandemic pressures.

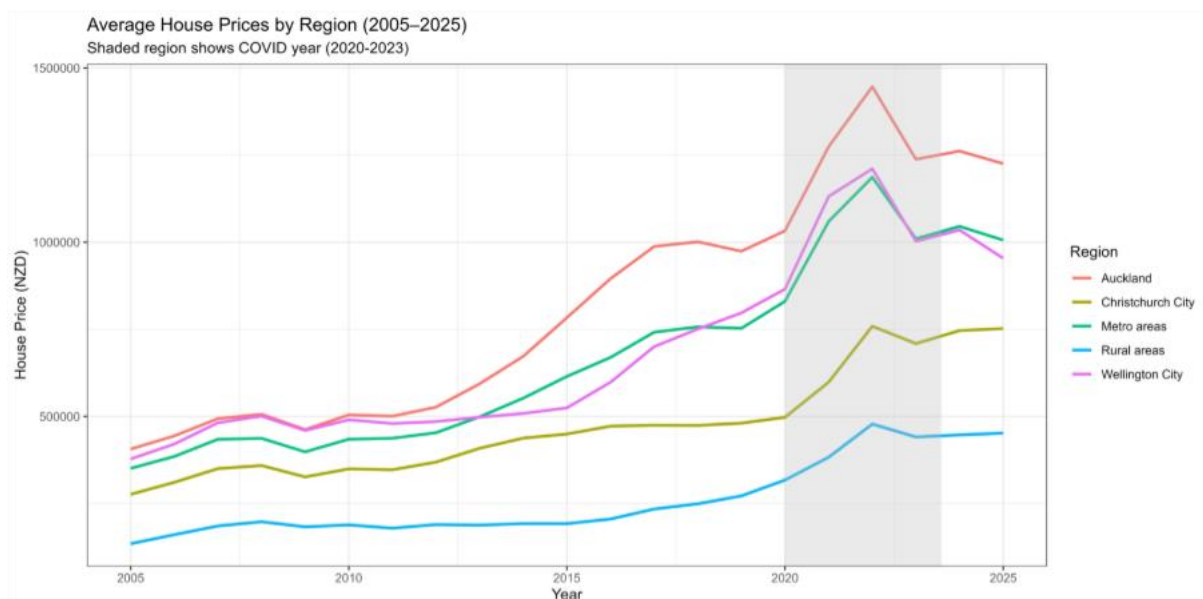


Figure 7. Yearly Average House Price by Region (2005–2025)

File: Yearly average house price by region.png

This chart showed nominal house prices rising across all regions, accelerating into 2020–2022 and easing after 2022. Auckland remained the most expensive level-wise throughout. The post-2022 cooling did not unwind the full earlier run-up.

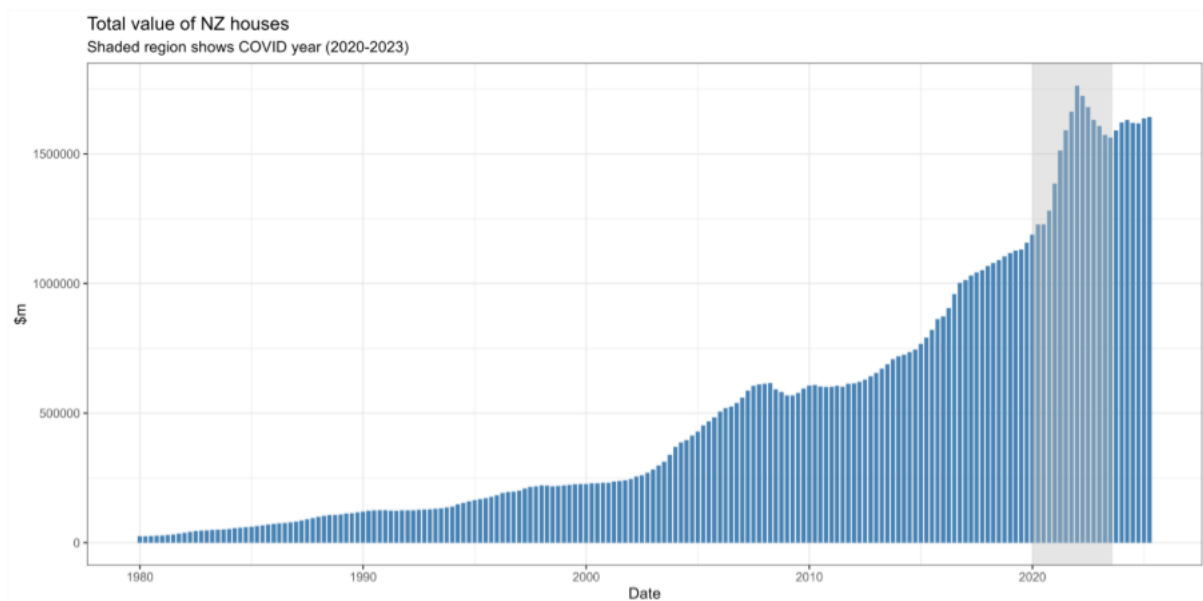


Figure 8. Total Value of NZ Houses (1979–2025)

File: Total value of NZ houses.png

This long-run series illustrated a steady increase in total residential value with a steep climb into 2021–2022 and a mild pullback thereafter. The broader trend remained upward despite the recent correction.

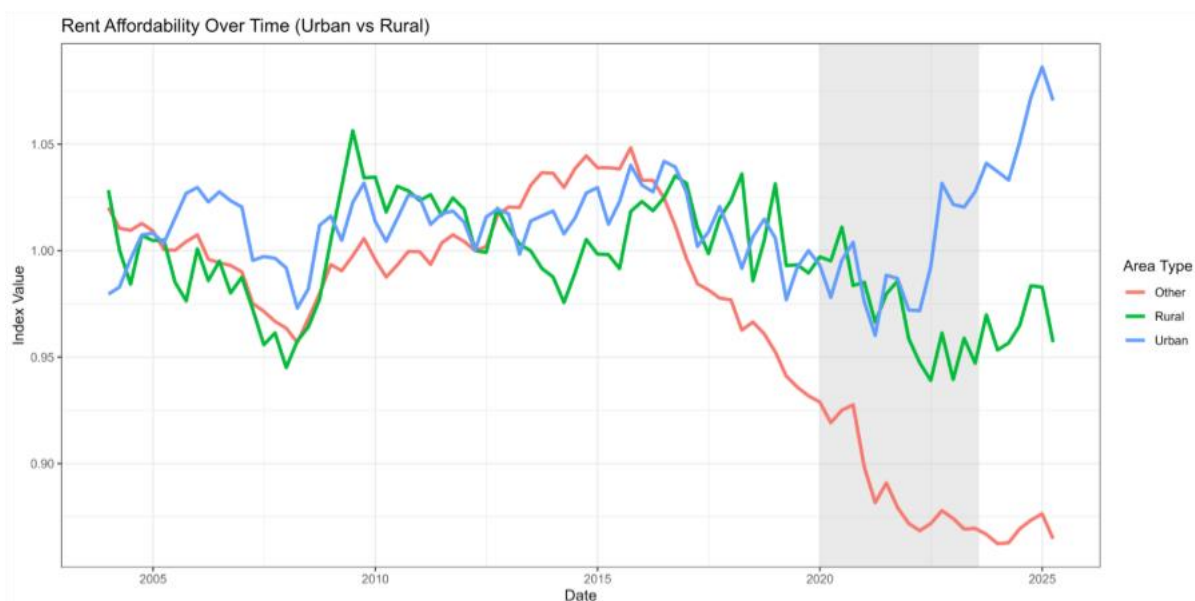


Figure 9. Rent Affordability Over Time (Urban vs Rural)

File: Rent_Affordability_Type_Urban_Rural.png

This figure contrasted rent affordability between urban and rural areas. Urban affordability worsened more from 2021 onward, widening the gap with rural areas during and soon after COVID. The gap narrowed only slightly by 2025.

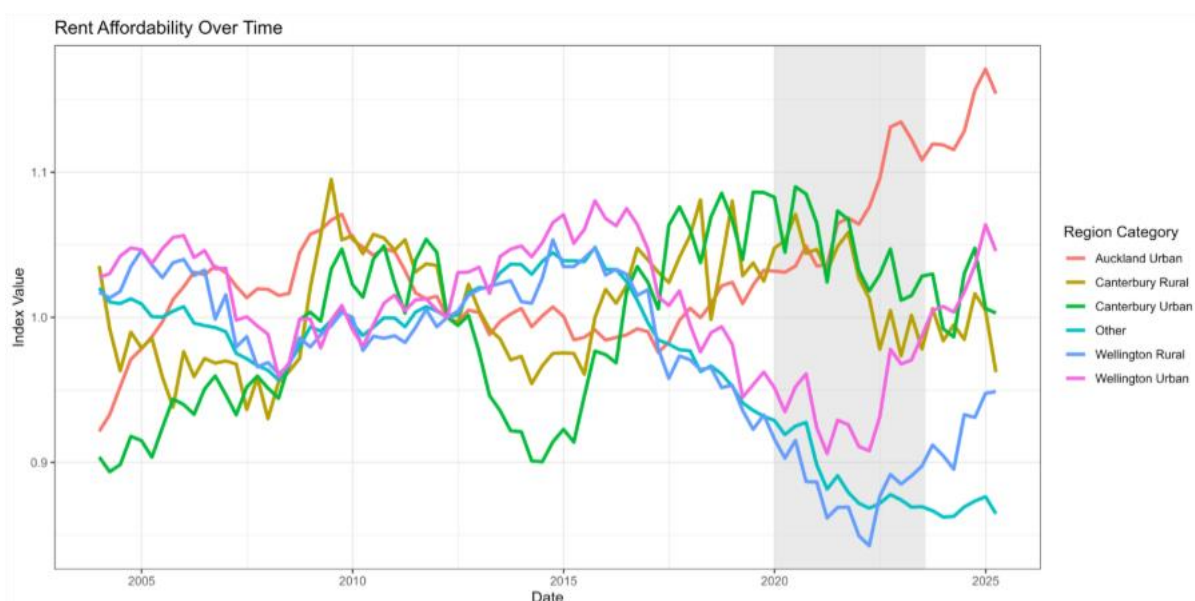


Figure 10. Rent Affordability Over Time (Detailed Region Categories)

File: Rent_Affordability.png

Here we broke rent affordability out by finer regional categories. Auckland-Urban trended less affordable after 2020, while Wellington categories dipped around 2020–2022 and then stabilised. The picture reinforced the urban pressure visible in Figure 9.

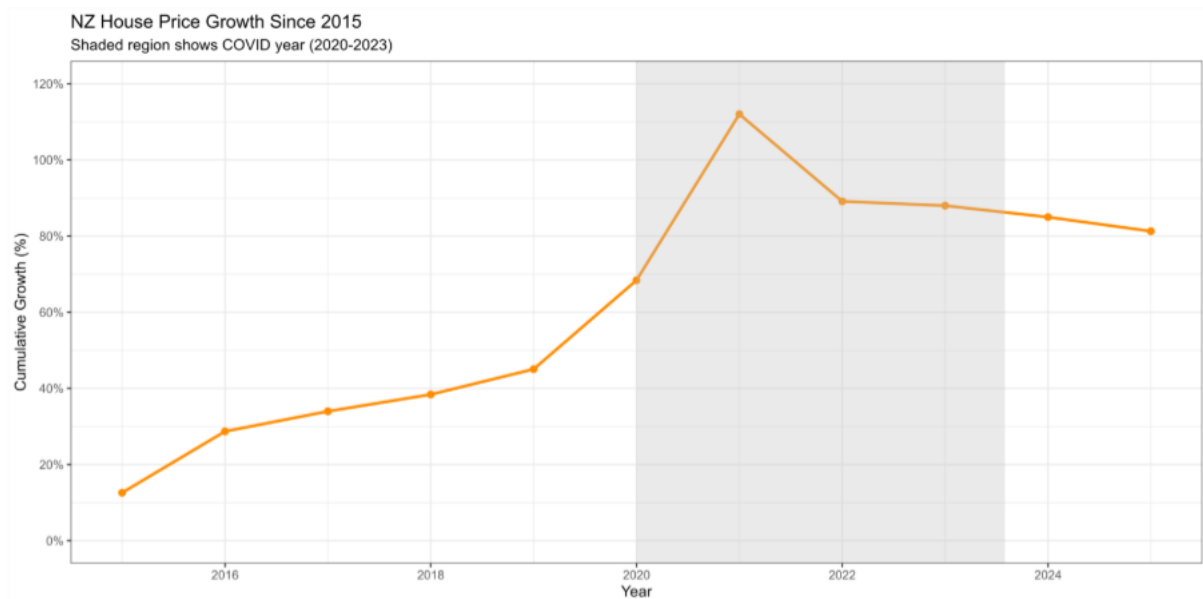


Figure 11. NZ House Price Growth Since 2015 (Cumulative)

File: NZ House Price Growth Since 2015.png

This cumulative index showed prices rising quickly through 2021, then partially reversing by 2023–2025. Even after the pullback, cumulative growth stayed well above 2015 levels, underscoring how large the earlier upswing had been.

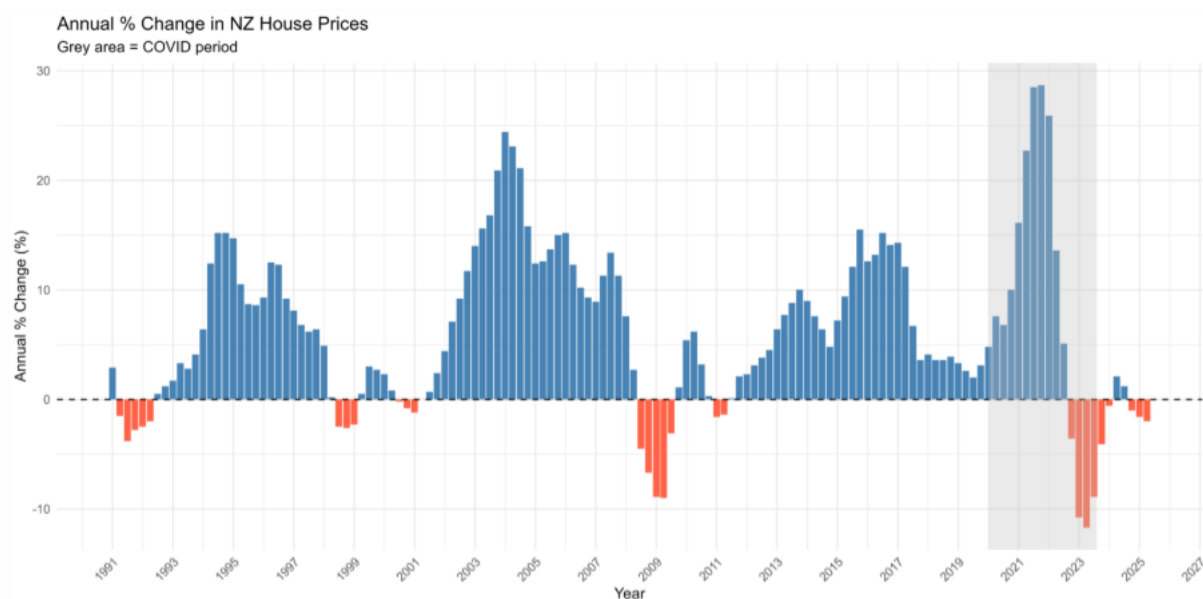


Figure 12. Annual % Change in House Prices (RBNZ series)

This growth-rate chart captured very strong positive price changes in 2021, followed by large negatives in 2022–2023. The pattern depicted a clear boom-then-bust in annual growth, consistent with the level charts.

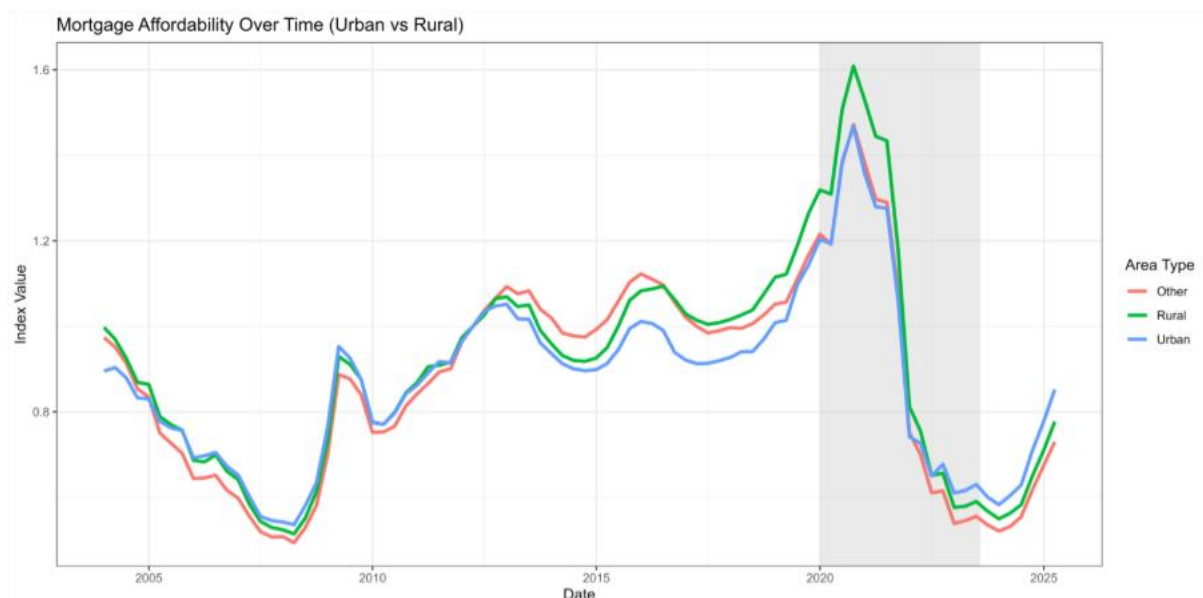


Figure 13. Mortgage Affordability Over Time (Urban vs Rural)

File: Mortgage_Affordability_Urban_Rural.png

This series tracked our mortgage affordability measure across urban and rural areas. Affordability tightened markedly in 2022 as rates rose, with a modest improvement in 2024–2025. Urban and rural moved largely together, mirroring the broader interest-rate cycle.

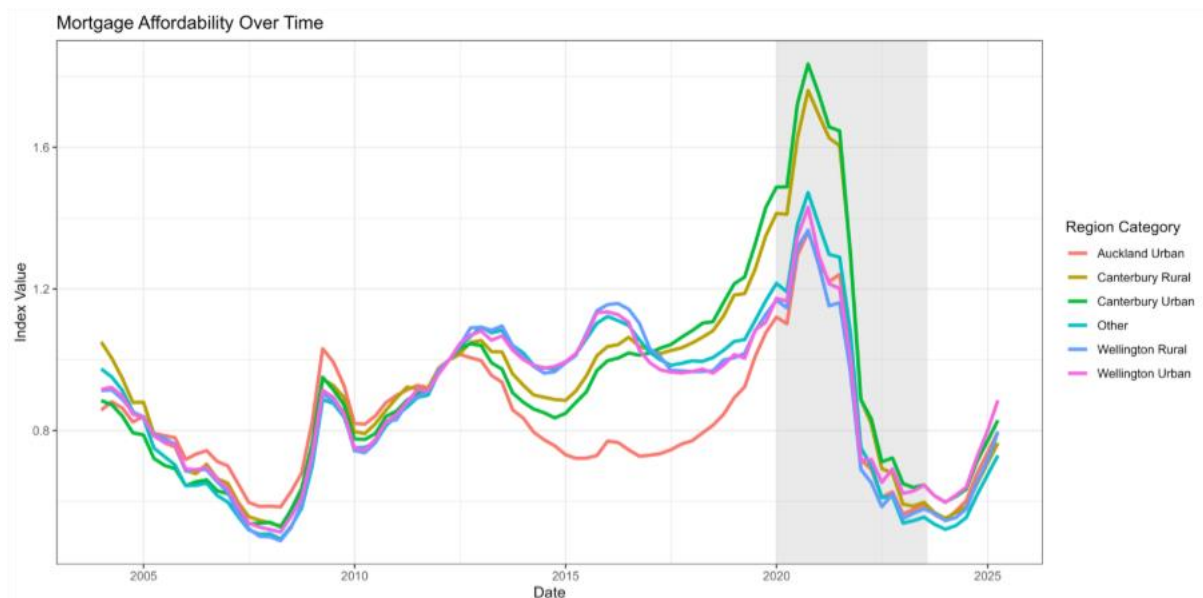


Figure 14. Mortgage Affordability Over Time (Detailed Regions)

File: Mortgage_Affordability.png

With more regional detail, this figure showed Canterbury-Urban experiencing the sharpest COVID-era swing. As rates eased slightly in 2024–2025, regions reconverged, although not back to pre-COVID conditions.

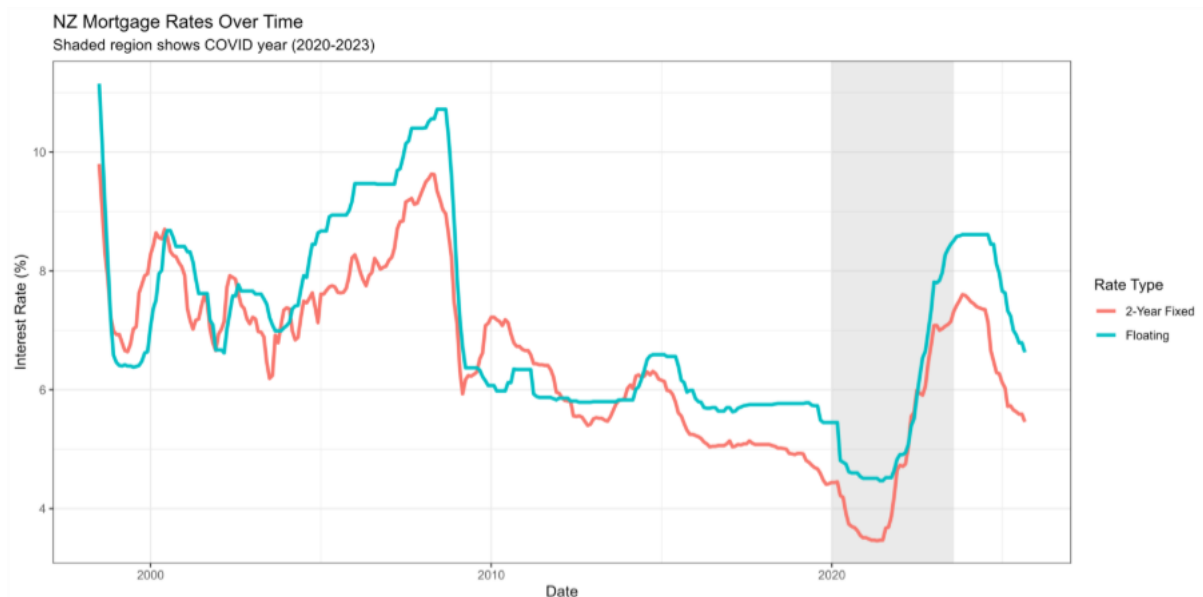


Figure 15. NZ Mortgage Rates: Fixed vs Floating

File: Fixed vs. Floating Morgage Rates.png

This plot displayed both floating and two-year fixed mortgage rates. Rates fell to about 3–4% in 2020, surged to around 7–8% by 2023, then edged down. Floating typically peaked slightly above the two-year fixed series.



Figure 16. Year-over-Year Change in Mortgage Rates

File: Fixed vs. Floating Morgage Rates YoY.png

This year-over-year change highlighted how abrupt the tightening was in 2022 and how 2024 brought net declines. The rapid swings matched the affordability shifts observed in our indices.

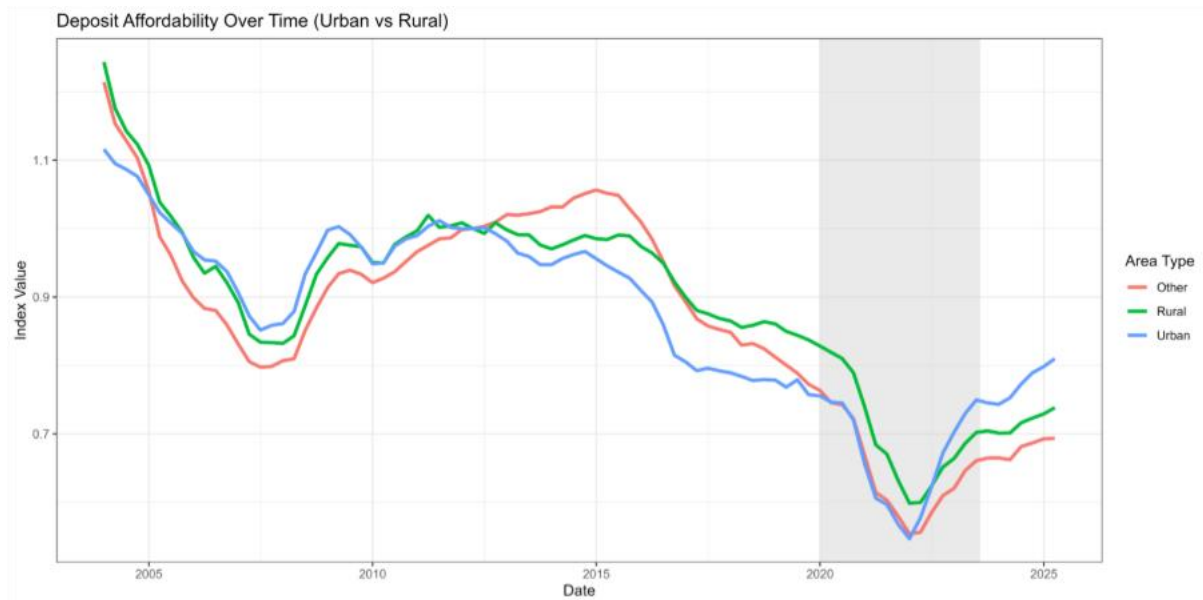


Figure 17. Deposit Affordability Over Time (Urban vs Rural)

File: Deposit_Affordability_Urban_Rural.png

Deposit affordability deteriorated into 2021–2022 and improved from 2023. Urban areas recovered a little faster, though the overall level remained more strained than pre-COVID.

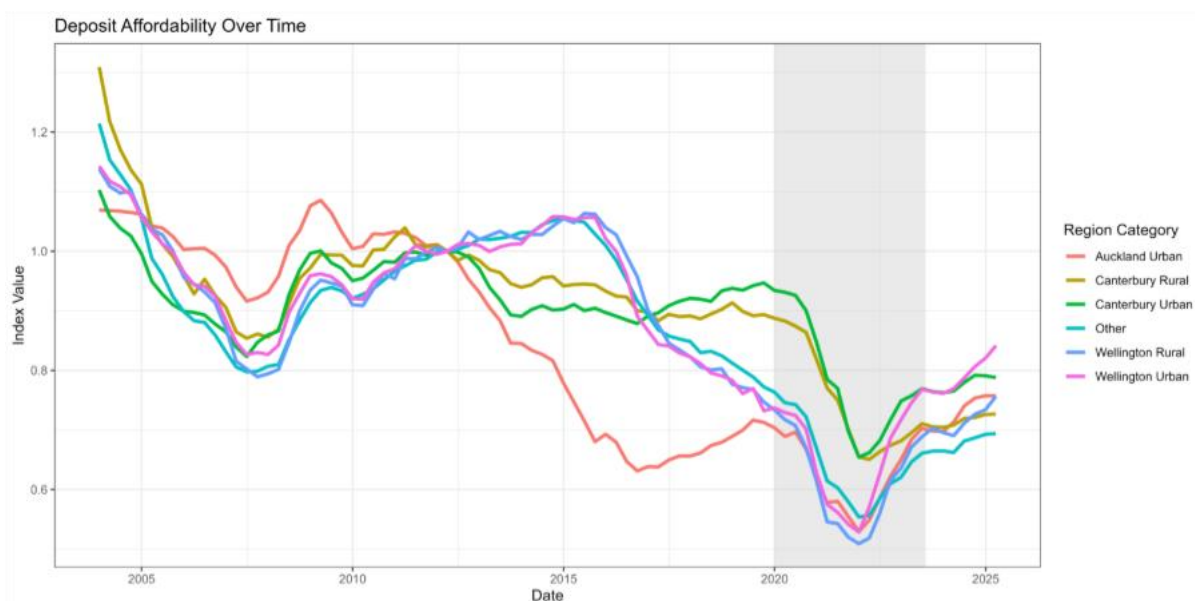


Figure 18. Deposit Affordability Over Time (Detailed Regions)

File: Deposit_Affordability.png

Across regions, deposit affordability worsened during COVID and then gradually improved. Auckland lagged in that recovery, consistent with higher entry costs.

Figure 19. Affordability Indices Over Time (Deposit, Mortgage, Rent)

File: Affordability Indices Over Time.png

This combined view compared deposit, mortgage, and rent affordability. Mortgage affordability worsened most sharply in 2022 with rising rates, deposit affordability fell into 2021–2022 and then rebounded, and rent affordability moved more steadily. Together, the indices traced the main pressures households faced through the cycle.

4.2. Model: Document the best model found and justify this choice. Quantify the best model in terms of its expected performance on unseen data.

Overview

We fitted three price models and two affordability models. We reported heteroskedasticity-robust (HC1) standard errors for the fixed-effects (FE) specifications and checked collinearity with VIFs. Below we summarised fit, key coefficients, and how each model contributed to the overall story.

Price models

(P1) Region fixed-effects price model (m2)

This model explained **94.5%** of the variation in log house prices (Adjusted $R^2 = 0.944$; residual SE $\approx$ **0.057** on 891 df). All core drivers were statistically significant with HC1 SEs: unemployment ($t=-22.8$), deposit index ($t=-9.15$), mortgage index ($t=-3.80$), rent index ($t=3.52$), and a strong time trend via Year_c ($t=17.34$). Region contrasts were large and precise: relative to Canterbury, Auckland's level was higher by **0.762 log points** (~114%); relative to Other and Wellington, Auckland was higher by **0.302** (~35%) and **0.281** (~33%), respectively.

(P2) Mixed-effects price model (m1)

The random-intercept model (1|Region) reached substantively similar conclusions. It detected a sizeable region intercept variance (SD $\approx$ **0.314**), consistent with real between-region differences after conditioning on covariates. Fixed-effect signs and magnitudes aligned with m2.

(P3) Urban–Rural price model (m3)

Focusing on Urban vs Rural (dropping “Other”), this model also fit well (Adjusted $R^2 = 0.923$; residual SE ≈ 0.087 on 201 df). We found a small but significant **urban premium** of **0.0365 log points** ($p=0.0156$) after controls. Mortgage and deposit indices were significant and negative; the time trend was negative in this restricted sample.

Affordability models

(A1) HAI fixed-effects model (m_aff_fe)

Using HAI = 1/Mortgage Index, the FE model detected a **post-COVID drop** in affordability ($\beta=-15.08$, $p=0.000011$) and a positive association with income ($\beta=41.33$, $p=0.0026$). Overall fit was modest (Adjusted $R^2 = 0.128$; residual SE ≈ 6.22), reflecting the proxy nature of HAI and the reduced sample once lagged consents were included.

(A2) Log price-to-income model (m_pi_fe)

As a robustness measure of affordability, the FE model for $\log(\text{price}/\text{income})$ fit strongly (Adjusted $R^2 = 0.952$; residual SE ≈ 0.213 on 430 df). Higher incomes **improved** affordability ($\beta=-1.45$, $p<0.001$), more lagged consents **improved** affordability ($\beta=-0.289$, $p<0.001$), and higher unemployment **worsened** it ($\beta=0.138$, $p<0.001$). The post-COVID indicator was collinear and dropped (singularity), which we noted explicitly.

Diagnostics and inference safeguards

VIFs indicated high collinearity for **Income_Level** and **Year_c** in several FE models (e.g., m2 Income VIF ≈ 42.5 ; Year_c ≈ 38.2). We therefore (i) centred year, (ii) reported **HC1** robust SEs for m2, m_aff_fe, and m_pi_fe, and (iii) interpreted those specific coefficients with caution while emphasising patterns that were stable across specifications (e.g., income $\downarrow$ log_PI; consents_lag1 $\downarrow$ log_PI). The HAI mixed-effects model flagged a **singular fit** (region variance ≈ 0), so we relied on **fixed effects** for HAI comparisons.

Expected performance on unseen data

We did not run out-of-sample validation. Based on in-sample fit, the price FE model (m2) and the log price-to-income FE model (m_pi_fe) explained a **very high share** of variation (Adjusted $R^2 \approx 0.94$ – 0.95) with small residual errors (≈ 0.06 in log-price; ≈ 0.21 in log price-to-income). These results suggested that the models captured the principal drivers in our dataset, while we cautioned that generalisation beyond the observed period and regions would require an explicit hold-out or cross-validation step.

Best model (highlight)

For **affordability**, the best-performing model was the **log price-to-income fixed-effects model (m_pi_fe)**. It achieved the highest explanatory power (Adjusted $R^2 = 0.952$) with coherent signs: higher incomes and more (lagged) consents improved affordability, while higher unemployment worsened it.

For **prices**, the strongest model was the **region fixed-effects price model (m2)** (Adjusted $R^2 = 0.944$), which explained nearly all variation in log prices and produced precise region contrasts.

Summary of model performance

Target	Model	Spec (key terms)	N	Adj. R ²	Residual SE	Notes
Price (log)	m2 (FE)	Region + Income + Unemp + Mortgage + Deposit + Rent + Year_c	901	0.994	0.057	Unemp ↓, Deposit ↓, Mortgage ↓, Rent ↑, strong time trend; large Auckland premium vs others
Price (log)	m1 (RE)	Same covariates + (1 Region)	901	-	0.057	Similar signs; sizeable region random-intercept SD ≈ 0.314 (between-region level differences)
Price (log)	m3 (FE, U/R)	Urban/Rural + covariates + Year_c	209	0.923	0.087	Small but sig. Urban premium (≈ 0.0365 log points) after controls
Affordability (HAI)	m_aff_fe (FE)	Region + Income + Unemp + Consents_lag1 + Year_c + postCOVID	235	0.128	6.22	Post-COVID drop (p≈2.2e-05); income positive; overall fit modest (proxy nature, reduced N)

Affordability (log price/income)	m_pi_fe (FE)	Region + Income + Unemp + Consents_lag1 + Year_c (+ postCOVID dropped as singular)	437	0.952	0.213	Income ↓ (better affordability), Consents_lag1 ↓, Unemp ↑; strongest overall fit for affordability
----------------------------------	--------------	--	-----	-------	-------	--

4.3. Method: Discuss why this particular ML algorithm might be particularly suited to your data and problem.

In our case, the problem is a short panel: few groups (regions = Auckland, Canterbury, Wellington, Other) observed yearly across 2015–2025, with questions about the drivers of, and differences in, affordability across regions/urban–rural. The structure is well-matched to linear fixed-effects (FE) regressions and a simple mixed-effects (random-intercept) sensitivity check.

First, the inclusion of Region fixed effects implicitly accounts for all time-invariant regional characteristics (planning regimes, geography, “base” amenities) without the need for measurement; that obviates omitted-variable bias and makes FE a natural fit to our stated goal of examining how drivers compare within each region over time. FE is also statistically robust to our relatively small sample sizes ($n \approx 901$ for prices; $n \approx 437$ for log price-to-income), where more-flexible ML models would overfit and would not yield the transparent, policy-relevant coefficients we seek.

Second, our outcome and covariates have clear economic-linear relationships after simple transformations (e.g. log price, log price-to-income), which stabilize variance and allow us to interpret effects as approximate percentage changes that are communicable to policymakers. We also include a one-year lag of a building-supply proxy (net building consents per 1,000, lagged within region) to account for timing, and we centre Year to reduce collinearity with the intercept. In cases where collinearity is

still high (Income, Year), we report HC1 robust standard errors to ensure valid inference.

Third, we also estimated a random-intercept HAI model, which was singular with only four groups: This indicated that, after accounting for controls, there was little remaining between-region variation to estimate, and limited identifiability. That finding reinforces our choice of Region FE as the preferred specification for affordability, and we view the mixed model only as a sensitivity check.

Finally, alternative approaches that may be “ML” in the narrow sense (random forests, boosting, deep nets) are not well-suited to our objective of explanation over prediction, given the short panel and need for coefficients associated with interpretable drivers (income, unemployment, rates, building supply). Pure time-series (ARIMA/VAR) or quasi-experimental approaches (e.g. difference-in-differences) would also be underpowered given our limited variation at the regional level and short time horizon. In short, FE linear regression (with robust SEs and appropriate data transforms) is the right tool for balancing interpretability, statistical robustness, and relevance to our policy questions.

5. Conclusions/Discussion

5.1. Fit-for-purpose: Discuss whether the results are statistically meaningful and fit-for-purpose. Relate your results to the original goals.

Statistical meaning (are the results solid?)

Prices. The region fixed-effects model (m2) explained **94.4%** of the variation in log prices (Adj. $R^2 = 0.944$). Using HC1 robust errors, key drivers were statistically significant: unemployment (large negative t), deposit index (negative), mortgage index (negative), rent index (positive), and a strong time trend. Region gaps were large and precise: Auckland’s level was much higher than Canterbury, “Other,” and Wellington. The urban–rural model (m3) also fit well (Adj. $R^2 = 0.923$) and showed a small but significant urban premium.

Affordability. The HAI fixed-effects model (m_aff_fe) detected a clear **post-COVID drop** in affordability and a positive link with income, but overall fit was modest (Adj. $R^2 = 0.128$), reflecting HAI's proxy nature and reduced sample. The **log price-to-income model (m_pi_fe)** performed strongly (Adj. $R^2 = 0.952$): higher incomes and more lagged consents improved affordability; higher unemployment worsened it. The HAI random-effects model was singular, so fixed effects were preferred for HAI comparisons.

Did we answer the original goals?

Drivers. Yes. Tighter financing (higher mortgage/deposit indices) and weaker labour markets (higher unemployment) were linked to lower prices and worse affordability; higher incomes improved affordability.

Regional and urban/rural differences. Yes. Auckland remained markedly more expensive even after controls; urban prices were slightly higher than rural.

COVID-19 impact. Yes. The post-COVID indicator and time-series plots showed an affordability squeeze in 2020–2023.

Is the approach fit-for-purpose?

Yes. Fixed effects controlled for time-invariant regional differences; HC1 robust errors supported inference; we tested urban/rural contrasts and used a supply lag. We diagnosed multicollinearity (high VIFs for Income and Year_c), centred Year, and interpreted those coefficients cautiously while relying on consistent signs across models.

Key limitations (what to keep in mind)

1. **HAI is a proxy**, which helps explain its lower fit.
2. **Pre-2019 consents gaps** reduced sample size in affordability models.
3. **Only four regions** limited random-effects identification for HAI.

5.2. Future work: Things that someone continuing your work should do next.

Future work would first address the **frequency mismatch** that limited several models. We would build consistent quarterly series for variables currently published annually (notably income and population) by transparently scaling from official indices. This would place all measures on a unified quarterly panel and allow like-for-like estimation across time.

Second, we would **automate data ingestion** to remove fragile manual steps highlighted in the constraints. A scripted pipeline would fetch and validate house-price files, normalise dates to YYYY–Qn, enforce a shared schema, and run reproducible checks (e.g., join tests and variable dictionaries) before analysis.

Third, we would improve the **affordability metric** itself. Because our HAI was a proxy constructed from a cost index, we would move to a standard qualification-income HAI that combines median income, prevailing mortgage rates (fixed and floating), loan term, deposit share, and practical DSR/DTI settings, reported with sensitivity bands. This directly addresses the lower explanatory power we observed for HAI.

On the **supply side**, the missing consents prior to 2019 reduced usable sample. We would test longer lags and incorporate completions where available to better approximate the pipeline from consent to delivery. This should strengthen identification of supply effects and mitigate the data-loss problem.

Methodologically, we would tighten identification and validation. To **absorb national shocks** that affect all regions, we would add two-way fixed effects (Region and Quarter). We would **quantify out-of-sample performance** using rolling-origin cross-validation and report MAE/RMSE on held-out quarters. To **stabilise estimates** in the presence of multicollinearity between income and time, we would include ridge as a robustness check and explore PCA on correlated affordability indices.

Given evidence that relationships may differ by place, we would **model heterogeneity** more explicitly, for example by allowing random slopes on rate variables (e.g., an OCR–affordability gradient that varies by region). Where timing and data permit, we would also run **event-study / difference-in-differences** analyses around major policy changes (e.g., LVR adjustments) to separate policy effects from background trends.

Finally, we would package the analysis as a **reproducible R workflow** (e.g., targets or make) so that data updates, re-estimation, and figure regeneration are one-click, supporting easy hand-over and auditability.

5.3. Problems: Discuss how problems that were encountered and project constraints were overcome.

Quarterly alignment

We aimed to analyse everything quarterly so results were comparable across charts and models. Several inputs (e.g., population and some income series) were only available annually. Rather than mix frequencies, we restricted modelling to variables with reliable quarterly coverage and flagged the frequency gap as a limitation. This choice kept the panel coherent, even though it reduced the set of variables we could include together.

To maintain transparency, we documented where quarterly series were genuine observations versus transformed inputs, and we deferred any annual→quarterly scaling to future work where it can be done with a clear, auditable method.

Income coverage

Median household income was essential for the price-to-income metric but was hardest to source consistently at the regional–quarter level. We consolidated the most complete series we could obtain for 2015–2025 and aligned its geography and timing with our panel. Where coverage fell short, we did not impute; instead, we excluded affected rows from the specific models that required income. This avoided injecting unverified values and kept inferences defensible.

House prices (historical depth and access)

OneRoof published monthly house prices, but historical CSV access for 7–8 years was not provided as a single download. We assembled the series month-by-month, validated totals against independent aggregates (RBNZ/Stats NZ), and then converted to quarterly averages for consistency. The process was time-consuming but protected data quality. We recorded the steps so this can be automated in a pipeline later.

Date formats and joins

Source files arrived with mixed date formats (e.g., “YYYY-MM”, “YYYY/Qn”, and text months). During the merge we normalised everything to a single calendar key (YYYY–Qn) before joining. This reduced silent merge failures and made downstream plots reproducible. We also applied consistent region names (Auckland, Canterbury, Wellington, Other) to prevent accidental duplication.

Missingness in supply (building consents)

Building consents were sparse prior to 2019, which sharply cut the usable sample whenever consents (or lagged consents) were included. We handled this with model-specific listwise deletion and reported the resulting sample sizes explicitly. We did **not** impute consents because the missingness reflected true lack of historical availability, not random gaps. In practice, this meant consents featured more in robustness checks than in the main models.

Multicollinearity and model stability

Income and time (Year) moved together, creating high VIFs in several specifications. We centred Year to reduce alignment with the intercept, reported HC1 robust standard errors for key fixed-effects models, and interpreted highly collinear coefficients cautiously. We prioritised results that were stable across specifications (e.g., income improving log price-to-income; higher rates tightening affordability).

Random-effects identifiability

With only four regions, the HAI mixed-effects model reported a singular fit (near-zero between-region variance). We therefore preferred region fixed effects for affordability comparisons and treated the random-effects results as supportive but secondary.

Tooling and reproducibility

Early work involved manual downloads and ad-hoc scripts. To reduce fragility, we standardised folder structure, enforced a shared schema (column names, types, and keys), and saved key intermediate tables used by both plots and models. We also version-controlled code and figures so that edits were traceable.

5.4. Ethics: Discuss any bias that might persist and the implications for how this might affect people.

Data source bias

Most of our data came from official agencies such as Stats NZ, the RBNZ, HUI, Stats NZ, etc. which we treated as credible and impartial. However, these sources still reflect how statistics are collected — for instance, they aggregate by broad region and may overlook local housing or income variation. This bias could slightly under- or over-state affordability for certain communities, especially smaller or rural areas that are averaged into a regional total.

Measurement bias

Our Housing Affordability Index (HAI) was a proxy, constructed from cost-side indices rather than true household-level qualification data. Because of this, it cannot perfectly represent the lived affordability experience of every household. The same applies to the price-to-income ratio, which uses median values that hide the wide spread of incomes and housing costs. These simplifications mean some social and economic realities remain invisible in our quantitative results.

Representational fairness

The regional and urban–rural comparisons we reported are descriptive, not prescriptive. It is important that our findings are not misused to justify reduced support or investment in less-affluent regions. High prices or low affordability are not indicators of success — they can also signal exclusion. Communicating these nuances clearly is an ethical responsibility, so that our results inform, rather than reinforce, inequality.

Privacy and transparency

All data used were public, aggregated, and anonymised. No personal information was accessed, stored, or linked across datasets. Nonetheless, we maintained transparency in how data were merged, transformed, and analysed, documenting each step to ensure reproducibility. This openness reduces the risk of hidden manipulation or accidental misrepresentation.

Responsible interpretation

Finally, we recognised that statistical relationships — such as the link between OCR and affordability — do not prove direct causation. Misinterpreting these associations could lead to misleading policy conclusions. We therefore presented all findings as correlations within the available evidence, clearly stating model limitations. Responsible interpretation ensures that the analysis supports informed debate rather than over-confident claims.

6. References: In MLA or APA style (or any recognized style). All copied text should be identifiable as such and attributed to a reference. Depending on the type of project we might expect a thorough literature survey to be reflected here.

Data sources

Ministry of Housing and Urban Development (HUD). (2015–2025). *Housing affordability indicators (HAM/HUD indices)* . <https://www.hud.govt.nz/>

OneRoof. (2017–2025). *OneRoof house price index and monthly value series*.
<https://www.oneroof.co.nz/>

Reserve Bank of New Zealand (RBNZ). (2018–2025). *Official Cash Rate (OCR)*.
<https://www.rbnz.govt.nz/>

Reserve Bank of New Zealand (RBNZ). (1998–2025). *Mortgage interest rates: Floating and 2-year fixed*. <https://www.rbnz.govt.nz/>

Reserve Bank of New Zealand (RBNZ). (2000–2025). *House price/value indexes & household debt-servicing ratios*. <https://www.rbnz.govt.nz/>

Stats NZ. (2018–2025). *Average (median) household income, annual series*.
<https://www.stats.govt.nz/>

Stats NZ. (2018–2025). *Building consents—dwellings (regional totals)*.
<https://www.stats.govt.nz/>

Stats NZ. (2018–2025). *CPI rental price index / rent payments*.
<https://www.stats.govt.nz/>

Stats NZ. (2018–2025). *Dwelling and household estimates*. <https://www.stats.govt.nz/>

Infometrics. (2010–2025). *Regional economic profile (house prices, incomes, unemployment)*. <https://www.infometrics.co.nz/>

Stephens, D. (2022, September 9). *Housing affordability in Aotearoa New Zealand: The importance of urban land supply, interest rates, and tax*.
<https://www.treasury.govt.nz/sites/default/files/2022-09/sp-housing-affordability-aotearoa-new-zealand-9sep22.pdf>

Wilson, N. (2020). *Fixing the housing crisis: The role of intergenerational policy design in addressing the issues*. **Policy Quarterly**, 16(2), 20–23.

7. Appendices: This is where any detailed material can go. For example notebooks of code, snippets of code, data dictionaries, supporting visualizations.

Table A1.

Variable	Number of Outliers	Range of Outliers
Income_Level	0	None
Unemployment_Rate	1	3.15
Mortgage_Index	4	3.12-3.50
Deposit_Index	2	2.86-3.05
Rent_Index	7	-2.97 - 3.67
Consents_Per_1000	346	None
Log_Price	0	None

Table A2.

Source Name	Variables Kept
Data.govt.nz	Mortgage_Index, Deposit_Index, Rent_Index, Mortgage_Rate, HPI Housing_Stock
Figure.nz	Unemployment Rates
Infometrics.co.nz	Average House Price
Rbnz.govt.nz	Debt_Servicing, CPI, House_Price_Change
Stats.nz	OCR, Rent_price, Building_Consents, Population

Figure A1.

Missingness Plot

